START HERE

How to read this pack

The 3Pin Realty platform is one connected system of five consoles that carries a property from the first seller message to a live post and a closed deal. This pack explains it at four depths.

5
consoles on one sign-in

24
connections between modules

10
CRM pipeline stages

9
Shoot & Media columns

4
scheduled jobs a day



One property, one path: the orange token is a listing moving from the first message to a closed deal.

Document	Answers	Read it if you are
1 · Functional overview	What does each module and process do?	Anyone new to the platform
2 · Business guide	How is each feature used, and why at that point?	Agents, shooters, editors, the owner
3 · Process & field reference	Every process, every field, every link between modules	Process owners, trainers, auditors, developers
4 · Architecture & flows	How the parts connect	Owner, developers, anyone changing the system
Release status & gaps	What went live, what is still to do, what is missing	Owner, release manager

Every statement is taken from the code itself: the live `main` at `b48bfaa` (released 13 Oct 2026, Firestore ruleset `132b5f73`), revised 14 Oct 2026. Work in progress in the working copy is listed under Release status and is not described as live.

DOCUMENT 1

Functional overview

What every console and process does, in one line each.

The platform runs the whole property business in five consoles that share one sign-in, one navigation rail and one database. A property moves left to right through them: it is **found** (CRM), **listed and shot** (Shoot & Media), **described and stored** (Property dashboard), **posted** (Posting) and **sold and billed** (CRM and Finance).

The five consoles

Console	Page	What it does, in one line
Home	`index.html`	Launcher tiles into every console, plus the customer-facing TailorTalk chat widget
Property dashboard	`dashboard.html`, `property.html`	The inventory: every property's facts, prices, links, map, matching buyers and data-quality worklists
CRM	`crm.html`	Every enquiry from first message to Won or Lost: pipeline, follow-ups, site visits, team calendar, AI placement, reports
Shoot & Media board	`propertytrack.html`	Every seller's property from "owner said yes" to "live online": details, documents, brochure, shoot, edits, voice-over, posting
Finance	`3pinfinance.html`	The agency's own double-entry books: deals, invoices, bills, GST, bank reconciliation, reports for the CA

Processes by module

Property dashboard

Process	What it does
Sync from the Inventory sheet	Copies property facts typed in the Google Sheet into the dashboard, protecting any edits made in the dashboard
Brochure pipeline	Turns a brochure request into a PDF, emails it to the team and adds or updates the property
Search, filters and client brief	Finds properties by natural language, 20+ filters, or a client's own words
Property detail	Shows facts, specs, sales pitch, notes, internal notes, matching buyers and a read-only **Shoot & Media and posts** block (the board card, its column and shoot, every post with its link)
Edit, sold out, interest, notes, delete	Keeps the working state of each property current; every edit is logged; a delete first says how many board cards and CRM leads still point at the property and keeps a full copy in the change log
Changes to apply	Lists dashboard edits so they can be copied back into the sheet by hand
Missing data, duplicates, area map	Data-quality worklists
Map	Positions, nearby places and travel times
Share	Sends plain details or the brochure to a client, or a login-only link to a colleague

CRM

Process	What it does
Lead capture	Creates leads from TailorTalk chats, the Add Lead form or a board card; never duplicates a phone number
Pipeline	Moves each lead along New → Options sent → Send more details → Visit planned → Visited → Negotiating → Won (plus Sourcing, On hold, Lost); **Leads for property** (`crm.html?propertyId=` , by id or code) shows only the leads linked to one property
AI lead automation	Reads each chat and places the lead, sets its next step and follow-up, books visits, links properties
Follow-ups	Schedules when the team contacts each lead and reminds them
Site visits	Books visits with Google Calendar invitations to the chosen agents and collects their answers
Team calendar and meetings	Shows everyone's day; finds free time; books meetings; a repeating meeting is edited or dragged one occurrence at a time
Daily task	One "your day" screen: visits, meetings, calls due, @mentions, voice-overs
Attention rules	Flags every lead that needs a person, by severity
Matching	Ranks properties for a buyer, and buyers for a seller
Reports	Analytics tab, 09:00 follow-up digest, 21:30 day summary, Excel export

Shoot & Media board

Process	What it does
Seller → card	Gives every seller lead a card automatically
Mapping	Links a card to its inventory property by hand (one property, one card; one card per lead)
Moving cards	Drag onto any part of a column, or the **Move to…** button on every card (works on phones)
Listing type and documents	Records Normal / Exclusive / Collab and the paper checklist (Have it / Applied for / Missing / Not needed); a record only — it gates nothing and never alerts
Brochure	Requests the brochure from the card and tracks it to delivery; reminds you to make a new version when the shoot came after the brochure
Shoot	Books, reschedules, runs (Shoots run sheet) and completes the shoot with the raw photos link; optionally puts the booked shoot on the team calendar (Settings switch, off by default)
Edits per outlet	Plans, starts and finishes one edit per outlet (Insta Story, Insta Reel, FB Reel, YouTube, Collab), with an optional final video link per cut
Voice-over	Asks for a V/O on the rough cut and routes it to the voice-over person
Media ready	Moves the card automatically once photos, edits, V/O and brochure are all done
Posting	Plans the week, schedules each channel, confirms posts live with their URL, tracks 99acres and the website
Configurations	Admin settings: editors, voice-over person, targets, tags, documents list, calendar switch

Finance

Process	What it does
Record	~35 plain-language actions that write balanced journal entries
Deal cycle	Token → registration → invoice each side → payment → settlement
Costs	Expenses, bills, payments, petty cash, salaries, assets
Recurring, loans, month-end	Monthly services, EMIs, depreciation and prepaid release
Bank reconciliation	Imports statements and matches them to the books
GST and TDS	Splits, credit, set-off order, returns registers
Reports and books	P&L, balance sheet, trial balance, cash flow, exports for the CA
Digest	08:00 email: weekly, renewals, monthly P&L

Automations at a glance

When	What runs
Every TailorTalk message	Lead created or updated; AI read 2 min after the chat goes quiet
Every save of a lead	The lead's row in `leadViews` is written in the same batch, so the CRM opens with about 39 reads instead of one per lead (760)
Every 5 min with the CRM open	Safety-net chat pull (resumes where the last one stopped) and due AI reads
Every Mac run	Brochure delivery, then Inventory sheet sync
04:30 IST	TailorTalk nightly sync
08:00 IST	Finance digest
09:00 IST	Follow-up digest and lead-views rebuild
21:30 IST	End-of-day summary email

DOCUMENT 2

Business guide

How each feature is used, and why at that point of the lifecycle.

Every feature exists to remove one of four business risks: **a lead nobody answers, a seller nobody lists, a property nobody can find the facts for, and money nobody tracks.** This guide follows the business day and the property lifecycle, and for each feature says when to use it, how, and what it saves.

The property lifecycle, stage by stage

Stage	Use this	How	Why at this point
1 · Find the owner	CRM (TailorTalk chats, Add Lead)	Owner enquiries arrive from WhatsApp, Instagram, the website, or are typed in	No owner is lost: the AI re-tags confident sellers, and a phone number is never entered twice
2 · Card for the property	Shoot & Media → Sellers / Board	A card appears for every seller automatically; **Create now** for any still waiting	"A seller who never gets a card is revenue that silently evaporates" — the Sellers badge shows who is still waiting
3 · Onboard	Card panel → Listing type, asking price, location; Edit details	Record Normal / Exclusive / Collab and the terms discussed	One place for the agreed terms, shown on the card and in the property's Shoot & Media block (the type is recorded; it gates nothing yet)
4 · Details & documents	Card → Listing & documents; brochure panel	Tick each paper Have it / Applied for / Missing / Not needed; fill title, photos folder, description, internal notes	The paper chase is visible per property ("n of m in hand"); nothing waits on it, so marketing is not blocked
5 · Brochure v1	Brochure panel → Generate	Send the request; in ~30 min the PDF is emailed and the card links itself	A brochure can go out before the shoot, from the photos already available
6 · Map to inventory	Card → Map to a property	Pick the property code by hand	Ties the card, the seller lead, Posting and the dashboard to one code — by hand so a chat can never mis-link it
7 · Plan the shoot	Card → Getting there, Book the shoot, Owner told	Address, contact, access notes; date, time, shooter; tick Owner told	The shooter arrives with everything; a clash within 2 h is flagged; "not ready to send" warns before a wasted trip. With the Settings switch on, the booking also lands on the team calendar with the shooter invited (off by default, because it sends real Google invitations)
8 · Shoot day	Shoots run sheet	Next up with **Call**, **Directions**, **✓ Done**	One phone screen for the whole day — no calls back to the office
9 · After the shoot	**Shoot done** dialog	Tick what was captured, paste the raw photos Drive link, add notes	Easy to update the shoot details right after completion; the card moves on by itself
10 · Plan posts and edits	Card → Shoot for; Plan the edit; Posting → Paste week plan	Choose outlets and V/O per outlet; assign the editor and deliver-by date; paste the owner's week plan	Every outlet's cut has an owner and a deadline; the plan text becomes scheduled rows
11 · Edit	Editor view	Plan → **Start editing** → **Edit done** per outlet; paste the **final video link** if you want it on the card	Each editor sees only their queue, late items first; time per edit is measured; the finished file can be found from the card
12 · Voice-over	**Ask for V/O** on the rough cut → **V/O made ✓**	The request lands in the voice-over person's CRM Daily task	No V/O is forgotten; it is asked for only when there is a cut to voice
13 · Brochure v2	Brochure panel → Redo brochure	Regenerate with the shoot photos	When a shoot is completed after the brochure was made, the card says so ("it does not have these photos") until a new version is made or **Not needed** is pressed
14 · Media ready	Automatic	The card moves itself once photos, edits, V/O and brochure are done	Nobody has to remember to move it; the column is always true

Stage	Use this	How	Why at this point
15 · Post	Posting → schedule → **Confirm posted** + URL	Schedule each channel; when it goes out, confirm and paste the link	Nothing is "posted" by the clock alone; every live post has its URL saved against the property and is listed in the property's **Shoot & Media and posts** block
16 · Market	Property dashboard → Matching Buyers; CRM → Properties for this buyer	Call the strongest matches first	The same engine everywhere, so every agent sees the same ranking
17 · Site visits	CRM lead → Site visit	Pick day, time and agents; save	Easily co-ordinate site visits: agents get a private calendar invitation with the client, seller and address; declines flag at once
18 · Negotiate and close	CRM stages; Finance deal	Move to Negotiating / Won; open a Finance deal, record the token, invoice each side	Brokerage, GST and receipts are tracked per property and side
19 · Sold	Property dashboard → Mark Sold Out; board → Sold / Rented	One tick takes it out of every buyer's matches	No agent pitches a sold property again

A day for each role

Sales agent

When	Do this	Why
Morning	Read the **09:00 digest** email; open **Daily task**	One list of today's visits, calls due and teammates' asks
All day	Work the **Follow-ups** → **Needs a person** queue, red first	Critical items (promise overdue, WhatsApp window closing) are worth the most
After every call	Answer **"What happened on the call?"** and set the next follow-up	Reminders: the next contact is never left to memory
On a buyer	**Properties for this buyer**, then + Link what you shared	Matching and a record of what was already sent
Before a visit	Schedule the **site visit** with the agents going	Everyone gets the invitation and the seller's number
After a visit	**They visited** → "How did the visit go?"	Moves the lead and captures feedback while it is fresh
On a lost lead	Pick the **reason**	Loss reasons feed the evening summary and analytics

Owner / Swami

When	Do this	Why
Weekly	**Paste week plan** in Posting	The plan becomes dated rows for every channel
On a card	Decide outlets and **V/O yes / separately** in Shoot for	Editors and the V/O person get their work from it
Evening	Read the **21:30 summary**	Escalations, the day's movement and why leads were lost
Any time	Board → **Needs attention** sort	Late cards (amber, red) rise to the top
On a property	Property dashboard → **Shoot & Media and posts**	Where the property stands in marketing — card, column, shoot, every post and link — without leaving the dashboard

Shooter

Open **Shoots** on the phone (tick "this is me" once). Work **Next up**: Call, Directions, ✓ Done. In **Shoot done**, tick what was captured and paste the Drive folder — that is the hand-off to the editor.

Editor

Open the **Editor** view (it opens on your own work). Late cards come first. **Start editing** when you begin, **Edit done** when finished (then **+ Final video link** if you want the finished file findable from the card — optional), and **Ask for V/O** once a rough cut exists.

Inventory maintainer

Type facts in the **Inventory sheet**; press **Sync from sheet** if the Mac is off. Work **Changes to apply → Pending**: copy each change into the sheet, then tick it. Clear **Missing data** before client calls; resolve **Duplicates** by hand.

Accounts / owner (Finance)

Use **Record** for every money event the day it happens (photograph the receipt). Import the **bank statement** monthly, reconcile, then **Run month-end**. Hand the CA the exports from **Books**.

Why it is built this way

Design choice	Business reason
Board mapping by hand only	A chat that mentions a code must never attach the wrong property to a seller
One property, one card	No two cards compete for the same shoot or brochure
Sheet is the typing surface, no automatic write-back	One place to type facts; edits made in the dashboard are listed for reconciliation, never lost
AI never marks Won and never overrides a person	Sales outcomes and team decisions stay human
Site-visit invitations only to team members, marked private	Client and seller phone numbers never leave the team
Documents never block posting	Marketing starts while papers are chased
Confirm posted needs a person	The URL record is always real
Append-only change log and ledger	Every edit and every rupee can be traced and recovered
Every admin option in Settings → Configurations	People and rules change without a code change
A save waits for the server's list ("Still connecting")	A phone that opens on its old saved copy can never write that copy over a colleague's newer work
Leads are read from packed view documents	Opening the CRM costs about 39 reads instead of 760, so the daily database bill stays flat as the lead count grows; the leads themselves stay the source of truth
Shoot invitations are off until switched on	A calendar invitation is a real email to a real person; the switch is turned on deliberately in Settings

Planned: "Eåsy by 3 PIN"

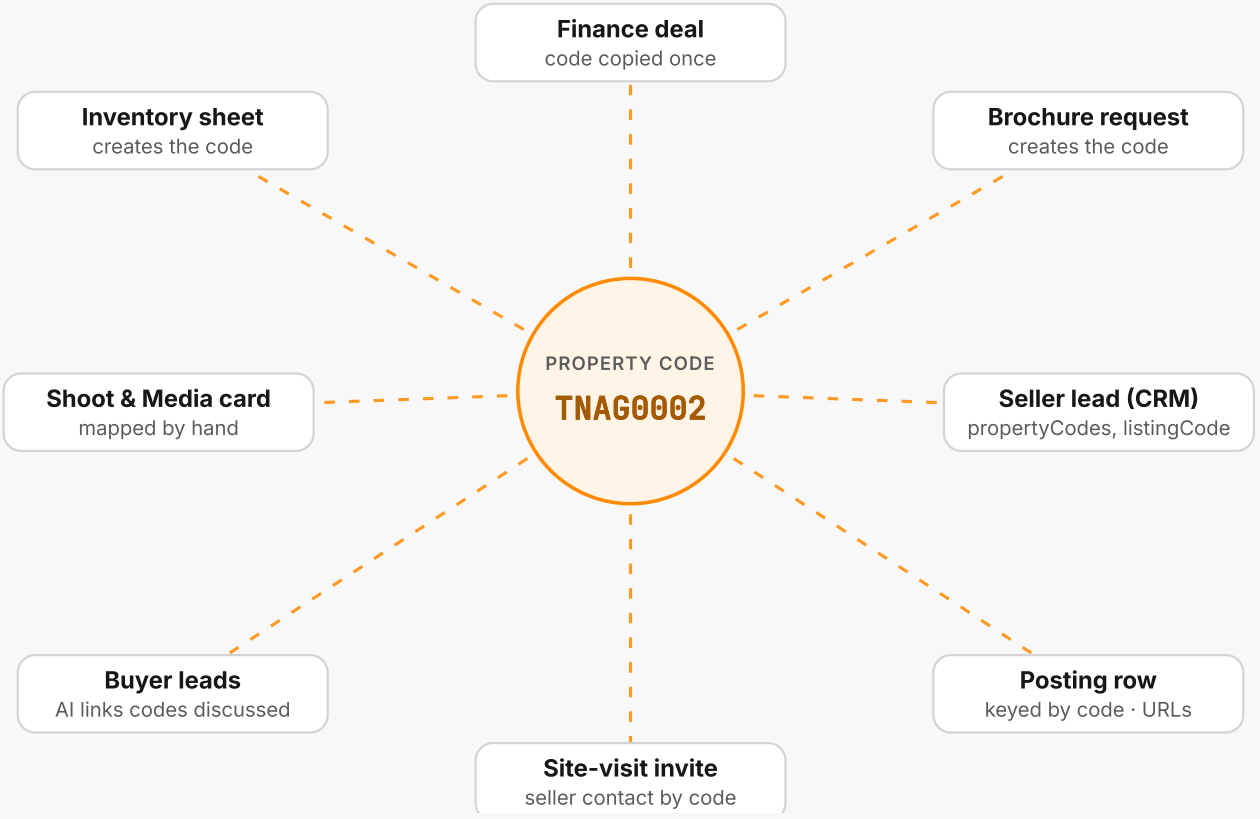
Eåsy by 3 PIN is the planned name for the platform's assistant and reminder layer — the digests, attention rules, AI placement and the nudges described above, presented to the team as one assistant with one voice. It is a product plan, not a live feature: nothing in the consoles carries the name today, and the reminders keep working exactly as described in this guide until it ships.

DOCUMENT 3

Process & field reference

Every process step, every field, and every link between modules.

The **property code** (e.g. `TNAG0002`: locality letters + serial number) is the thread that ties the modules together; the lead id and the listing id tie a person to a card.



The property code at the centre: every module reads or writes it. Pulses show data moving along each link.

3F · Every interconnection between modules

The **property code** (e.g. `TNAG0002`) is the thread that ties the modules together; the **lead id** and the **listing id** tie a person to a card. Every link below names the exact field that carries it and which way data flows.

The connection matrix

#	From → To	Carried by	Direction	When it happens	Rule
1	Customer chat → CRM lead	TailorTalk webhook / pull → `leads.tt`, name, interest, budget, enquiry type, conversation	In	Every message; nightly; every 5 min	A person's edit wins (`ttHold`)
2	CRM lead → AI → pipeline	`leads.ai`, `stageId`, `followUpAt`, site-visit fields	Both	2 min after a chat goes quiet; after notes	Forward only, never Won, never over a person
3	Chat → property links	AI finds codes → `leads.propertyCodes`	In	Each AI read	Only codes that exist; never re-adds an unlinked one; **never maps a board card**
4	Seller lead → Shoot & Media card	`listings.leadId`, `sellerName`, `sellerPhone` ↔ `leads.listingId`	Both	Board open, or sync script	Every seller lead gets a card unless set aside
5	Board card → new CRM lead	"Create the lead" → `leads` (source board)	Out	By hand	Duplicate phone links the existing lead
6	Board card → inventory property	`listings.propertyCode`	Board reads property	**By hand only**	One property, one card; mapping fills blanks; unmap takes back only what it fetched
7	Board mapping → CRM lead	`leads.propertyCodes`, `listingCode`	Board → CRM	On mapping	Old mapped code removed on re-map
8	CRM "Not a seller" → board	`enquiryType`, `listingSkipped`, card delete	CRM → board	By hand	Card removed only if untouched
9	Inventory sheet → property	Sheet sync → `properties` (tier A)	In	Mac run / Sync button	Pending dashboard edits protected
10	Dashboard edits → sheet	`propertyChanges` worklist	By hand	Maintainer ticks	No automatic write-back, by design
11	Brochure request → property	Form → Queue → Cowork → Mac → `brochureLink`, `photosLink`, `detailsText`	In	~30 min after request	Fills blanks on existing properties
12	Property → board brochure state	`brochureLink` → card `brochure.doneAt`, panel frozen	Dashboard → board	Within seconds of delivery	Only the card that asked for it
13	Property internal notes → board	`internalNotes` → card internal notes	Dashboard → board	On mapping	Fills a blank only
14	Board card ↔ Posting row	`propertyCode`, `reference`, edit gates, V/O tags, `finalLinks`	Both	Planning and posting	Schedule needs edit planned; live needs edit done; the final video link is optional
15	Posting → board	Confirm posted → card moves to **Live**	Posting → board	On Confirm posted	Forward only

#	From → To	Carried by	Direction	When it happens	Rule
16	Board / Posting → dashboard	`listings` + `postTracker` → "Shoot & Media and posts" block (by id or code)	Read only	When the block is opened	Shows each card's column, listing type and shoot, the seller lead, every post's status and URL; a delete warns about cards and leads still linked
17	Board → CRM Daily task	`forVoAsk`, `voiceByEmail` → "Voice-overs to make"	Board → CRM	Live	Marks "mine" for the voice-over person
18	CRM site visit → Google Calendar	Visit fields + property seller contact → private invitation	Out, RSVPs back	On save	Team agents only; stamped events
19	Board shoot → Google Calendar	`shootAt`, shooter → 60-minute event on the booker's calendar, `shootEventId`, `shootEventBy`	Out; moves, re-assignments and cancellations follow	On booking	Only with the Settings switch on (off by default); only the entry's owner may move it
20	Property ↔ matching	Match engine reads properties + leads + cards	Read only	On open	Same scores in all three consoles; sold out = ruled out
21	Dashboard → CRM filter	`crm.html?propertyId=` (Lead CRM button)	Link	By hand	Shows only leads linked to that property, matched by id or code
22	Property → Finance deal	`propertyCode`, `propertyName` copied once	Read once	Add a deal	Later renames not reflected
23	CRM Won → Finance	–	**None**	–	Deals and invoices entered by hand
24	Leads → digest / EOD email	Lead views + attention rules	Out	09:00, 21:30	Same rules as the board
25	Every lead write → `leadViews` row	`leads/{id}` + `leadViews/{tenant}__v{n}_{i}.rows[id]` (`_w` server stamp)	Same batch or transaction	Every save from any console or server job; rebuilt daily at 09:00	All or nothing; a page writes one row per document per write; `scripts/lead-views.mjs off` is the kill switch
26	CRM seller lead → board card	`leads.listingId` → `propertytrack.html?listing=` ("Open its card")	Link	By hand	A seller without a card gets "Track this listing" instead

How the property code travels

1. A **brochure request** or **Inventory sheet row** creates the code (`properties.propertyCode`).
2. An agent **maps** a Shoot & Media card to it by hand → `listings.propertyCode` .
3. The board copies it to the seller's lead → `leads.propertyCodes` , `listingCode` .
4. The AI adds it to **buyer** leads that discussed the property → `leads.propertyCodes` .
5. Posting rows are keyed by it → `postTracker.propertyCode` ; post URLs are saved there.
6. A **site visit** on that code pulls the seller's name, number and location into the calendar invitation.
7. A **Finance deal** copies it once.

8. The dashboard reads cards, posts and leads back by the same code.

3A · Property dashboard (Property Intelligence)

Console: `dashboard.html`, share page `property.html`. **Collections:** `properties` (+ `notes`, `internalNotes`), `propertyChanges`, `syncBackups`, `brochureLog`.

The Property dashboard is the agency's single shared list of properties for sale — builder projects and resale / owner listings. It answers an agent's questions: *what do we have, what exactly is it, what can I send the client now, who wants it, what fits this client, where is it, and is the data complete?*

Screens

Screen	Purpose
Property grid (List / Split / Map)	One card per property: code — name, location, agent, status badge, price, rate, area, resource chips (Brochure, Photos, Map Pin, Details), Call, View Details
Search box + Advanced Search	Natural search ("3 and 4 bhk in anna nagar under 1.5cr", "sqft:>2000", "builder:sobha") and 20+ filters with live counts
Client brief	Paste what the client said → the inventory is ranked for them (35 %+, top 40, with reasons for anything ruled out)
Map view	Price pins by precision, nearby places in 12 categories, properties within 5 km / 30 minutes, distance to a typed address; drop an exact pin
Property detail panel	Header actions (Share, Save, Lead CRM, Mark Sold Out, Print, Edit, Delete), agents, and tabs: Overview · Matching Buyers · Specifications · Sales Pitch · Notes & CRM · Internal Notes
Shoot & Media and posts (a block in the detail panel)	Read-only view of the property's marketing, read when the block is opened: each board card with its column, listing type (Exclusive / Collab) and shoot date, **Open card** → and **Seller lead** →; every post (Insta Story, Insta Reel, FB Reel, YouTube, 99 Acres, Website) with status, date and **Link** → or "no link", reposts marked. Changed only on the board and in Posting
Missing data	Properties with empty fields, worst first; type a value or mark **Not required**
Changes to apply	Every dashboard edit, from → to, who and when — tick once copied into the sheet; CSV export
Sync from sheet	Preview the sheet → dashboard changes, then apply all, selected or one
Area map	Audit of the Chennai geography the system has learned from the data
Duplicates	Pairs that look like one property entered twice — delete the spare by hand
Create brochure	Submit a brochure request (code assistant, photos folder, details, internal notes) and track recent brochures
Share page `property.html`	Fast one-property view for logged-in colleagues (Overview, Specifications, Sales Pitch only)

Process 1 — How a property gets in

Route	Who / what	Record id	Notes
Inventory sheet → Sync (recommended)	The Mac every run, or **Sync from sheet**	Sheet column A `Property_ID`	Facts are typed in the Inventory master sheet; the dashboard reads it
Brochure pipeline	Create brochure / Google Form → Queue sheet → Cowork PDF → Mac delivery	First word of the Queue title	Creates the property if new; on an existing one only fills blanks and sets the three delivery links
+ Add Property	An agent	`p<timestamp>` (even if a code is typed)	Lives only in the dashboard; never synced — the usual source of duplicates
First-login seed	System	"1"…"46"	Sample data, once per tenant

Process 2 — Sync from the Inventory sheet

1. Read the Inventory sheet and the Queue sheet (columns found by **header text**, never by letter).
2. Build each row into a property. Column F packs *type* | *sale type* | *stage* | *possession* | *age*; the stage becomes the two-value **status** (Ready to Move / Under Construction) and the original wording is kept.
3. Unknown headers go to **From the Inventory Sheet** (`sheetExtras`) automatically — a new column needs no code change.
4. **Empty cells never clear a value.** Queue photos and details only fill blanks.
5. **Protection:** any field with an un-ticked dashboard change is skipped, so the dashboard's value stays; a pending delete stops the property being recreated.
6. Old properties with no sheet row are **never touched**.
7. Preview first; apply writes a backup (`syncBackups`, or `backups/` on the Mac) and then merges.

Process 3 — Brochure pipeline (end to end)

1. **Submit** on Create brochure (or the Shoot & Media board's brochure panel): "CODE - Title", Drive photo folder, property details, internal notes. The code assistant suggests the next free code, flags existing codes and near-misses.
2. A row lands in the **Queue sheet**; the request is logged in `brochureLog`.
3. **Cowork** (Claude on the owner's Mac) builds the PDF and sets the row to *Done*.
4. The Mac's **deliver-brochures** job uploads the PDF to the photos folder in Drive, makes it link-readable, emails it to Thirumal, Swami and Pradeep ("Brochure ready: ID — title"), marks the Queue row *Brochure Emailed*, writes the link into the Inventory sheet's `Brochure_Link`, and creates / updates the property in the dashboard. Failures send a "[FAILED]" email (once per 24 h).
5. **Recent brochures** shows In queue / Generated — being sent / Delivered / Needs attention.
6. The Shoot & Media board sees the new property within seconds and links the card that asked for it.

Corrections: `scripts/replace-brochure.js CODE [--notify]` swaps the PDF everywhere.

Process 4 — Editing, sold out, interest, notes

Action	Where	What it does	Logged?
Edit	🖍️ Edit (26 fields, old → new shown inline)	Saves the property (merge); each changed field gets a change-log entry and is protected from the sync until ticked	Yes
Mark Sold Out	Header button	SOLD OUT overlay; **ruled out of matching for every buyer**; hidden from nearby; "sold" on CRM chips. The sync never un-sells	Yes
Interest level	Notes & CRM → 🔥 Hot / 🌡️ Warm / ❄️ Cold	Team-shared heat of interest (nothing else reads it yet)	Yes
Notes & Events	Notes & CRM	Note, Site Visit, Client Call, Price Update, Status Change, Booking — with a date	No (separate log)
Internal notes	🔒 tab	Team-only instructions (owner's number, negotiation floor, access). Never on the share page. Imported ones (Queue, Cowork) become manual once edited	No
Agents	Header dropdowns	Property agent and Secondary	No
Map pin	Map → Drop the exact pin	Exact position; survives syncs	No
Missing data	Missing data panel	Fill a value, or **Not required** (stored in `naFields`)	Yes
Delete	🗑️	Says how many board cards and CRM leads still point at it (they keep the code but no longer find the property); saves a full copy in the change log **first** — no copy, no delete — then deletes. If the delete itself fails the property is put back, the "Property deleted" entry is set aside as done and a line says the delete did not go through	Yes (with snapshot)

Saves wait for the server. A full save from the Edit form is refused with "Still connecting — your changes are kept here. Press Save again in a moment" until this device has received the server's property list, so a tab that opens on an hours-old local copy can never write it back over a Sync from sheet or a colleague's edit. Only partial saves — the exact map pin — go through before that.

Process 5 — Copy dashboard edits back to the sheet (by hand, by design)

There is **no automatic write-back**. The maintainer opens **Changes to apply → Pending**, makes the same change in the Inventory sheet, then ticks the row. Ticking hands that field back to the sheet: from the next sync, the sheet's value wins. Entries can never be deleted.

Process 6 — Matching buyers and client briefs

The **match engine** (shared with the CRM and the board) scores every buyer lead against the property in four layers: rule-outs (wrong type, a stated objection, sold out) → weighted attributes (Budget 28, Location 22, Configuration 16, Type 14, Size 10, Possession 9, Lifestyle 8, Amenities 6, Vastu 5, Approval 4, Facing 4, Builder 4, and smaller ones) → wording similarity → objection penalties. Bands: **85 %+ Strong match · 70 %+ Worth a call · 55 %+ Partial · below that Long shot**. Matching Buyers lists them with Open in CRM and Call.

Process 7 — Sharing

With	How	What they get
A colleague	🔗 Share Link	`property.html?id=…` — needs a login
The CRM	**Lead CRM** header button	`crm.html?propertyId=…` — the CRM filtered to the leads linked to this property, matched by its id or its code
A customer	💬 Details → Copy → WhatsApp	The exact intake description (`detailsText`), no internal link
A customer	📑 Brochure / 🖼️ Photos	The public PDF / the photo folder
Anyone	Print, JSON, Compare + CSV	Printable summary; raw record; side-by-side table

Field reference — property (`properties/{id}`)

Ownership tiers decide who may write each field: **A** = Inventory sheet (facts) · **B** = dashboard (working state) · **C** = brochure pipeline (delivery links) · **D** = system.

Field	Label	Values	Tier	Sheet column	Read elsewhere / why
`propertyCode` / `id`	Property Code	Locality letters + number (TNAG0002, MYLA002, LUX049)	A / D	`Property_ID`	**The key that joins every module** — CRM leads, board cards, Posting rows, Finance deals, calendar invites
`name`	Property Name	Text	A	`Property_Name_Project`	Everywhere; board title; Finance deal name
`builder`	Builder	Text (default Individual Owner)	A	–	Search; matching (Builder 4)
`location`, `zone`	Location, Zone	Narrow → broad text; region	A	`Location_Area`, `Zone`	Search, matching, map, site-visit invite "Where"
`type`, `saleType`	Type	Apartment, Villa, Plot, House…; New / Resale	A	Column F parts 1–2	Type rule-outs; New/Resale filter
`status`	Status	**Only** Ready to Move or Under Construction	A	Column F part 3	Counters and filters — any other value drops out of both
`constructionStage`, `possession`, `propertyAge`	Stage, Possession, Age	Text	A	Column F parts 3–5	Badge text; matching (Possession 9, Age 2)
`availability`	Availability	Available / Immediate / Sold	A	`Availability_Status`	Search filter (not a matching rule-out)
`config`	Configuration	"3 BHK", "2, 3 BHK"	A	Config column	Matching (Configuration 16)
`startingPrice`, `pricePerSqft`	Starting Price, Rate / Sqft	"₹1.25 Crores", "₹7999/Sqft"	A	`Price`, `Rate_per_Sqft_INR`	Matching (Budget 28), map price pills, board asking price, Finance picker
`sqftRange`, `superBuiltupArea`, `carpetArea`, `totalLandArea`, `uds`	Areas	Text sqft / cents / grounds	A	Area columns	Size matching; land search
`totalUnits`, `totalTowers`, `totalFloors`, `floorNo`	Project / building	Text	A	Matching columns	Specs; matching (Floor 3)
`facing`, `bathrooms`, `parking`, `parkingType`, `furnishing`, `cornerUnit`, `vastu`, `powerBackup`, `approval`	Key facts	Text	A	Matching columns	Filters and matching weights
`highlights`, `amenities`	Highlights, Amenities	Comma lists	A	`Highlights`, `Amenities`	Sales Pitch; 26 amenity tags
`nearbyLandmark`, `connectivity`, `mapLink`	Location & connectivity	Text; Maps link	A	Matching columns	Area model, map position
`contactName`, `contactNumber`	Contact	Text	A	`Owner_Builder_Contact`	Call buttons; **site-visit invitation "Seller: name · number"**

Field	Label	Values	Tier	Sheet column	Read elsewhere / why
`sheetNotes`, `sheetExtras`	Notes; From the Inventory Sheet	Text; header → value	A	`Notes`; any unmapped column	Search; new columns appear without code
`brochureLink`, `photosLink`, `detailsText`	Brochure, Photos, Shareable Details	Drive links; text	C	`Brochure_Link`, `Images_URL`; Queue C / D	Resource chips; board brochure state; client WhatsApp text
`soldOut`	Mark Sold Out	Yes / No	B	–	**Matching rule-out**, map, CRM chips, board
`interestLevel`	Client Interest Level	hot / warm / cold	B	–	Team signal
`naFields`	Not required	List of fields	B	–	Missing data count
`assignedAgent`, `secondaryAgent`	Property agent, Secondary	Team emails	B	–	Agent filter
`geo`, `mapPinSnoozed`	Map pin	lat, lng, precision	B	–	Distances on the map
`createdAt`, `updatedAt`, `source`, `tenantId`	–	–	D	–	Sorting, audit, scoping

Never write the eight dead names `propertyId`, `propertyType`, `price`, `priceInCr`, `readyToMove`, `newOrResale`, `possessionDate`, `builtupArea` — they once caused dashboard edits to silently revert.

Missing data checks 30 fields: core (config, area, price, rate, possession, availability), specs (land, UDS, units, floors, floor no., facing, bathrooms, parking, furnishing, vastu, power backup, approval, age), location (zone, map link, landmark, connectivity), marketing (highlights, amenities, photos, brochure, details) and contact (name, number).

Related records

Record	Contents	Why
`properties/{id}/notes`	Kind (note, visit, call, price, status, booking), date, text, author	Shared history of what happened with this property
`properties/{id}/internalNotes`	Text, source (manual / queue-sheet / cowork-generation)	Team-only instructions; the board reads them when mapping
`propertyChanges`	Field, from → to, who, when, applied-to-sheet tick, full copy on delete	Reconciliation worklist and sync protection
`brochureLog`	Title, who, when	Recent brochures status

3B · CRM — leads, follow-ups, site visits and calendar

Console: `crm.html`. **Main collections:** `leads` (+ `notes`, `history`, `tailortalk/state`, `aiRuns`), `pipelines/{tenant}`, `settings/{tenant}`, `leadViews`.

The CRM holds every enquiry — buyers and tenants looking for a property, owners who want to sell or rent out (the source of new listings), and vendors or collaborators kept apart. It does four jobs: run the **pipeline**, drive **daily execution** (follow-ups, site visits, mentions), turn **chats into leads** automatically, and **report** on all of it.

Screens

Screen	Where	What it is for
Board	CRM → Board	One column per stage; drag, Alt+←/→ or "→ next" to move; days-in-stage chip turns amber then red
List	CRM → List	Excel-style table, 14 columns, column filters, bulk Move / Set follow-up / Clear follow-up
Follow-ups	CRM → Follow-ups	"Needs a person" queue (Do now / Today / This week) and the follow-up calendar (Overdue → Later)
Analytics	CRM → Analytics	Stat cards, trends, funnel by milestone, pipeline by stage, owner load, today's drill-downs, data hygiene
Daily task	Rail → Daily task	"Your day": site visits, meetings, calls due, @mentions asked of you, voice-overs to make
Team calendar	Rail → Team calendar	Everyone's day side by side, Find a time, New meeting, answer invitations. A repeating meeting is listed as single occurrences: the edit form says "Repeats — this change is for this one only", and one occurrence may be dragged to a new time (the series as a whole may not)
Lead panel	Click a lead	Everything about one lead (see below)
Alerts bell	Header	Unanswered invitations, today's items, leads that need a person

Lead panel, top to bottom: AI summary · Stage and Sent Details · Lead agent and Secondary · Site visit · Log a note (with @mentions and next follow-up) · Where this lead stands (milestones, next step, attention reasons, AI suggestion / AI moved with Undo) · Follow-up spotlight · Lead info (incl. linked Properties, each opening the dashboard, and on a seller the **Listing** row, which opens the property's card on the Shoot & Media board — or "Track this listing" when it has none yet) · Properties for this buyer / Buyers waiting (matching) · TailorTalk section (status, Overview, Conversation) · Timeline.

Navigation rules. The search box and the filter chips act on a list of leads, so they are hidden on Daily task, Team calendar and Analytics (Analytics counts every lead). `crm.html?propertyId=…` — from a property's **Lead CRM** button — shows only the leads linked to that property, matched by its id or its code, as a "🏠 Leads for CODE" chip with a count. A lead or card panel owns one history entry while open: **Back** (phone or browser) closes the top dialog first, then the panel, never the page; a lead opened from a link (`?lead=`) is not reopened once closed.

The pipeline

#	Stage	Meaning (rule shown under the column)	Target days
0	New	Nothing specific sent yet	2
1	Options sent	Sent a property, details or location; they are weighing it up	7
2	Send more details	They asked for something we have not sent yet	2
3	Visit planned	Visit asked for or agreed, not done yet	5
4	Visited	They have seen the property	4
5	Negotiating	Talking price, token or documents	14
6	Won	Token paid, signed or rented — **only a person sets this**	–
side	Sourcing	We know what they want and have nothing to show yet	3
side	On hold	Interested, but paused — reason + optional revisit date	–
side	Lost	Stopped — reason required	–

Lost reasons: not interested, bought elsewhere, budget, no match, unreachable, spam, not a fit, other.
Hold reasons: postponed, budget, other.

Every move records who moved it and when, and the first date each milestone was reached (`reached`) — this powers the funnel. Moving to Lost or Won clears the follow-up; a hold with a revisit date turns that date into the follow-up.

Process 1 — A lead arrives

Source	How	What happens
TailorTalk (WhatsApp, Instagram, website chat) — the main source	Webhook on every message, plus safety-net pulls (nightly 04:30 IST, every 5 min while the CRM is open — a pull that runs out of time resumes where it stopped — and the Sync TailorTalk button)	The lead is found by TailorTalk id, then by phone (`phoneKey` = 91XXXXXXXXXX), or created; the full conversation is mirrored; name, interest, budget and enquiry type are filled until a person edits them (`ttHold`); the default agent is assigned
Add Lead form	Channel, Name, Phone, Enquiry type, Property/Locality required	If the phone already exists, the Duplicate dialog offers Open or Update — a second lead is never created
Shoot & Media card	"Link a CRM lead → Not in the CRM? Create the lead"	A Seller Listing lead in New, linked to the card; a phone already in the CRM offers the existing lead instead

A person's edit always outranks TailorTalk: once a shared field is edited it is held, and the panel shows "TailorTalk now says X — use it?".

Process 2 — Assignment

Each lead has a **Lead agent** and a **Secondary** agent, chosen by a person (new leads get the default agent from settings). A newly assigned person gets an @mention-style task in "Asked of you", on the card and in their digest email. Site-visit agents are chosen separately per visit.

Process 3 — Follow-ups (when *we* contact the lead)

1. Set **Next follow-up** from the note row, the Log follow-up dialog, the Add/Edit form, a bulk action or an On hold revisit date. Presets: +2 h, +4 h, Tomorrow, +3 days, +1 week, +2 weeks. Times step in 30 minutes; a date alone means 10:00.
2. After tapping Call or WhatsApp, returning to the tab within 5 s–30 min asks **"What happened on the call?"**.
3. **✓ Followed Up** logs it and sets the next date (or none).
4. Reminders: the bell, the Follow-ups badge, optional browser notifications (every 15 min, at most hourly), Daily task "Calls due" and the **09:00 IST digest**.

Process 4 — Site visits (when everyone meets at the property)

1. **Schedule** on the lead: day, time, minutes (30/60/90/120), in person or by phone, **who is going** (with live free / busy / on leave from Google), property, notes for the agent, status (Asked for / Agreed / Done / Cancelled).
2. *Gates:* no date → status must be Asked for or Cancelled; Agreed with a date → at least one agent.
3. Saving sends a **private Google Calendar invitation** to the chosen team agents only (client and seller numbers never leave the team). The event carries client name and phone, what they want, budget, the seller's name and number from the property, the location and a link back to the lead.

4. Agents answer in Gmail/Calendar or in the CRM ("I can go / Can't make it" + reason). A decline raises **"Agent cannot make the site visit"** on the board, bell and digest.
5. Moving the time asks everyone again; **Call it off** deletes the event and tells everyone.
6. Three hours after the visit, **"Did the site visit happen?"** appears; **They visited** moves the lead to Visited and asks "How did the visit go?". After two silent days in Visited, "Get feedback on the visit" appears.

Process 5 — AI lead automation

When switched on (⋮ → Lead automation), an AI reads each lead after its chat goes quiet for 2 minutes (at most 10), after every note, or on **Re-check**. Default model Gemini 3.5 Flash-Lite; Claude Haiku can be switched in (capped at 100 reads per day).

The AI decides	Values	What the CRM does with it
Intent	buy, rent, sell, rent out, invest, vendor, unclear	A confident sell / rent-out re-tags the lead **Seller Listing** → a board card appears
Stage	one of the 10 keys + confidence + quoted evidence	Moves **forward only**, never to Won, never over a person's choice unless the lead wrote since; uncertain moves become a **suggestion** (Move / Dismiss); every AI move can be **undone** for 7 days
Next step	owner (team / lead), action, kind, due date	Sets the follow-up inside 09:30–19:30 IST; shown as the card's next-step line
Site visit	requested / scheduled / done / cancelled, time, property	Writes the visit fields unless a person set them after the lead's last message
Hold / lost reason	reason + revisit date	Lost only for not interested, bought elsewhere or spam at high confidence
Status line	≤ 80 characters	Shown on the card
Property codes mentioned	codes that exist in inventory	Added to the lead's linked properties

Phones and emails are masked before anything is sent to the model, and every read is logged (`aiRuns`).

A separate **AI summary** (Claude Haiku, on demand via Regenerate or Backfill) writes a 3–4 line summary from the lead's details, last 25 notes and 40 history lines.

Process 6 — What needs a person (attention rules)

One rule book drives the board, bell, queue, digest and browser alerts. A reason stays open until anyone on the team touches the lead.

Reason	Severity	When
Promise overdue	critical	The team's next step is past due
WhatsApp window closing	critical	We owe a reply and under 3 h remain of the 24 h window
Team owes a reply	high / medium	Our next step is open
Follow-up overdue	high	Follow-up time has passed
Waiting for team / escalated / flagged	high	TailorTalk handed the chat to a person
Visit declined	high	An agent declined a visit
Did the visit happen?	high	3 h after a planned visit
Details owed	high	In Send more details for over a day
Sourcing stale	high	In Sourcing for over 3 days
Feedback due / negotiation stalled / hold due	medium	Visited 2+ days silent; negotiating 3+ days silent; revisit date reached
Lead silent / stale / AI suggestion	low	Waiting on the lead 3+ days; untouched 21 days; AI suggests a move

Process 7 — Matching and property links

- **Properties for this buyer:** the match engine reads the requirement (BHK, type, budget, localities, size, possession, objections) and scores the whole inventory — 40 % or more is shown, top 40, with vetoed properties and reasons.
- **Buyers waiting:** on a seller lead, the same engine scores every open buyer against the owner's property.
- **＋ Link / Link to this lead** adds a property code to `propertyCodes`; a person's unlink is remembered and never re-added by the AI or the board.

There is **no brochure-sending button** in the CRM; details reach clients through TailorTalk or WhatsApp, and the CRM records the fact with **Sent Details**, the Options sent column and the linked properties.

Process 8 — Seller leads and the Shoot & Media board

A lead is a **seller** when its enquiry type is Seller Listing, or the AI reads sell / rent out with high confidence. The board then gives it a card automatically. The CRM owns the person (name, phone); the board owns the property. The property code is mapped **only by hand on the board** and then copied back to the lead. **Not a seller? Move back to leads** re-tags the lead, locks the type and removes the card only if nobody has worked on it.

Process 9 — Reports and emails

What	When (IST)	Content
Follow-up digest	09:00 daily (and Send Digest Now)	Needs action now, overdue, today, tomorrow, day after; each recipient's own @mentions. Email (WhatsApp delivery is configured but unreliable)
EOD dashboard summary	21:30 daily (and Send Now)	Escalations first, then the day's metrics; AI-phrased reasons for leads lost today
Export Leads	On demand	`.xlsx` with 25 columns and all notes

Field reference — lead (`leads/{id}`)

Field	Label	Values	Written by	Read by / why
`name` , `phone` , `email`	Name*, Phone*, Email	Text	Person, TailorTalk, board	Everywhere; invitations; digest
`phoneKey`	–	91XXXXXXXXXX	Every save	Dedupe across TailorTalk, form and board
`channel`	Channel*	call, whatsapp, instagram, website	Person, TailorTalk	How they reached us; analytics
`source`	Source badge	manual, tailortalk, board (legacy meta, whatsapp_bot)	Create path	Intake origin; analytics
`enquiryType`	Enquiry Type*	Property Enquiry, Seller Listing, General, Vendor / Collaboration, + custom	Person, TailorTalk, AI re-tag	Buyer / seller / vendor routing; seller → board card
`propertyInterest`	Property / Locality*	Text or "CODE · Name"	Person, TailorTalk, board	What they want; matching; invitation
`budget`	Budget	Text ("1.5 Cr")	Person, TailorTalk	Pipeline value; matching
`detailsSent`	Sent Details	Yes / No	Person, AI	Did we share details
`stageId` , `prevStageId` , `stageChangedAt/By` , `reached`	Stage, milestone trail	Stage ids, times	Person, AI	Column, stage age, funnel
`lostReason` , `holdReason` , `holdUntil`	Why lost / on hold, Revisit on	Enums, date	Person, AI	Loss analysis; revisit
`followUpAt` , `followUpBy` , `followUpSetAt` , `followUpNote`	Next follow-up	Time, who, why	Person, AI, TailorTalk booking	When we call next and why
`assignedAgent` , `secondaryAgent`	Lead agent, Secondary	Team emails	Person, default	Ownership; filters; digest
`mentions`	@ you	Per teammate: by, at, text, done	Person	Asks between teammates
`siteVisitAt` , `siteVisitStatus` , `siteVisitProperty` , `siteVisitAgents` , `siteVisitReplies` , `siteVisitNotes` , `siteVisitMinutes` , `siteVisitMode`	Site visit	Time; requested / scheduled / done / cancelled; agents; RSVPs	Person, AI, calendar	The viewing and who goes
`calendarEventId` , `calendarEventOwner`	–	Google ids	Calendar API only	Keeps the invitation in step
`propertyCodes` , `unlinkedPropertyIds`	Properties	Inventory codes	Person, AI, board	Lead ↔ property link; matching "already shared"
`listingId` , `listingCode` , `listingSkipped`	Listing	Card id, code, flag	Board, person	Seller ↔ board card
`tt` , `ttHold`	TailorTalk section	Snapshot; held fields	TailorTalk; person	Chat status, escalation, ad, signals
`ai`	Where this lead stands	Verdict, suggestion, last move	AI	Placement, next step, Undo
`aiSummary`	✨ AI summary	Text	AI summary	Quick read

Field	Label	Values	Written by	Read by / why
`lastNote`, `noteCount`, `lastActionType`	–	–	Person	Cheap reporting without extra reads
`createdAt/By`, `updatedAt/By`	Added / Updated	Time, who	Create / person only	Audit; `updatedAt` = "a person last worked this lead"

Sub-records: `notes` (text, by, mentions) · `history` (every change; by a person, AI, TailorTalk or a migration) · `tailortalk/state` (AI profile, bookings, payments, full chat) · `aiRuns` (every AI read, server only).

Lead views (`leadViews`) — how the CRM opens cheaply

Firestore bills one read per document, so a list of 760 leads used to cost 760 reads on every open. Since 13 Oct 2026 every lead also lives as a **row** inside a view document holding about 20 of them: `leadViews/{tenant}__meta` (layout: version, shard count, count, built time, `off` switch) and `leadViews/{tenant}__v{n}_{i}` (`rows: { leadId: lead + _w }`). The CRM, dashboard and board read the views — **39 reads instead of 760** — and then one read per change, as before.

- A row **is** the lead (same fields; notes and history stay in their sub-collections), so every screen reads it exactly as it read the lead.
- Every writer — CRM, board, dashboard, TailorTalk sync, AI, calendar — writes the lead and its row **in the same batch or transaction**, stamped with the server's time (`_w`). A page saves only the fields it changed, so a row that was behind can never carry an old value back into a lead.
- The **09:00 digest rebuilds** the views from the leads themselves, keeping any row written after the leads were read; when the leads outgrow the documents a larger set is prepared and switched to a day later.
- Switches: `node scripts/lead-views.mjs verify` (must print "IN STEP"), `build`, `off` (every page reads the leads directly again, within seconds — the kill switch), `on`.

Settings the CRM uses (`settings/{tenant}`)

Setting	Where set	Effect
Enquiry types, property names	Add Lead form	Dropdown options
Follow-up digest (WhatsApp / email, recipients)	⋮ → Follow-up Digest	09:00 digest
Dashboard summary email (recipients)	⋮ → Dashboard Summary Email	21:30 email
Lead automation (on/off, model)	⋮ menu; script for model	AI placement
Saved views	☆ Views	Team-wide filter sets
Team list	**Script only**	Who can be mentioned, invited, shown in the calendar
Default lead agent, TailorTalk workspace	**By hand in Firestore**	New-lead owner; TailorTalk links
Lead views on / off	**Script only** (`node scripts/lead-views.mjs off` / `on`)	Whether pages read the packed views or the `leads` collection directly

3C · Shoot & Media board and Posting

Console: `propertytrack.html` (formerly "Property & Media"). **Collections:** `listings`, `listings/{id}/history`, `trackPipelines/{tenant}`, `postTracker`, `settings/{tenant}.trackConfig`.

The CRM follows a *person* through a sale; this board follows a *property* from "the owner said yes" to "it is live in the market". Its founding rule, written in the code header: *a seller who never gets a card is revenue that silently evaporates* — so every CRM seller lead gets a card automatically.

Views and who uses them

View	Used by	Purpose
Board / List	Agents, owner	Every listing in its column, with days-in-column against a target. A card moves by drag — dropped anywhere on the column, not only on a card — or with the **Move to…** button every card carries (the way to move on a phone)
Shoots (run sheet)	Shooter	Today's and upcoming shoots with Call, Directions and Done buttons
Editor	Editors	One card per outlet cut: plan → editing → done → scheduled → posted
Sellers	Agents	Every seller lead as a tile, and sellers still waiting for a card
Posting	Owner / poster	The posting plan, week calendar, sheet of channels and post URLs
Settings → Configurations	Admin	Editors, voice-over person, targets, tags, documents list, calendar switch

The code has **no roles**: any signed-in team member can do everything. People are recognised by settings (editors list, voice-over person) and by a "this is me" switch on the shooter's phone.

Saves wait for the server. Until this device has received the server's list of cards, every change — a move, a dialog's Save, a delete — is refused with "Still connecting — this is the copy saved on this device. Try again in a moment" (a dialog stays open with what was typed). A phone that opens on an old local copy can therefore never write that copy over a colleague's newer work.

Board columns

Column	Kind	Target days	Rule shown under the header
New listing	open	2	Owner is on board — nothing collected yet
Details & docs	open	3	Chasing the write-up, price and papers
Shoot scheduled	open	5	Date fixed and the owner knows
Shoot done	open	2	Photos taken — upload, then generate the brochure
Media ready	open	3	Media, photos, voice-over and brochure all done
Live — marketing	open	30	Listed and being shown
Sold / Rented	won	–	Closed
On hold	hold	–	Reason and optional revisit date kept
Dropped	lost	–	Reason kept

Lateness = days since `stageChangedAt` against the target (Settings can override each). Past target the pill turns amber; past twice the target it turns red, and "Needs attention" sorting pushes the card up (+2 over target, +3 over double, +3 more if the shoot is within 24 hours or already past).

Process 1 — A card is created

1. **Automatically from a seller lead.** When the board is open, `reconcileSellers` gives a card to every lead that (a) is a seller — enquiry type *Seller Listing*, or the AI says intent *sell / rent out* with high confidence, (b) is not a business lead, (c) is not set aside, and (d) has no card yet. The card takes the owner's name and phone, a title from the lead's property interest and the asking price from the budget. **The property code is never filled from the lead.** One lead gets **one card**: a lead that already has a card is never given a second one, and linking a card to a lead that already has one is done only on purpose.
2. **Sellers view** → **Create now** for a seller still waiting.
3. **+ New listing** by hand (title or code required).
4. **From Posting** — Create brochure on a row with no card makes one (in Live if a post is live, else Media ready); a new series makes a series card.
5. **From the script** `scripts/sync-seller-listings.mjs` (same logic, run by hand; not scheduled).

To opt a seller out: **Set aside** (`listingSkipped`), **Delete listing** (also sets `listingSkipped`), or CRM **Not a seller** (deletes the card only if nobody has worked on it).

Process 2 — Card and lead stay in step

Direction	What moves	Rule
Lead → card	Seller name and phone	Only until someone edits them on the card (`own.*` then wins)
Lead → card (once)	Title, location, asking price	Copied at creation, then the board owns them
Card → lead	`listingId`, `propertyCodes`, `listingCode`	The card's code is added to the lead; the code the sync added before is removed on re-map; codes a person unlinked in the CRM are respected
Card → CRM (new)	A whole Seller Listing lead	"Link a CRM lead → Not in the CRM? Create the lead" (name + phone; a duplicate phone offers the existing lead)

Process 3 — Map the card to inventory (by hand only)

Mapping ties a board card to a property in the Property dashboard by its **property code**. It is done only by a person — Map to a property, the Property ID box, the brochure panel's Link, or Posting's Create brochure. One property can sit on **only one** card.

- **Linking** fills only empty card fields from the property (title, location, configuration, price, description, photos link, internal notes), always takes the delivered brochure link, and records exactly what it fetched (`fetched`).
- **Unmapping** takes back only what the mapping fetched. If a brochure was requested from the card, all details stay and only the code is removed.
- **The one automatic link:** when a brochure requested from this card under this code reaches the dashboard, the card is linked to it.

Typed in Property ID	Result
Empty, on a linked card	Unlinks
A code already in the dashboard and on another card	Refused
A code in the dashboard, free	Linked after saving
A code not yet in the dashboard	Kept as the brochure's draft code

Process 4 — Listing type and documents

- **Listing type** (`listingType`): Not set / Normal / Exclusive / Collab — a chip on the card, in the form and in the panel, and shown on the property's Shoot & Media block when Exclusive or Collab. It is recorded but **gates nothing yet**.
- **Documents checklist** (`docs`): Sale deed / title, Parent documents, Encumbrance certificate (EC), Patta, Approved plan, Property tax receipt, Owner ID & PAN — the list is edited in Settings → Configurations (remove every item to hide the checklist). Each is *Have it / Applied for / Missing / Not needed*; the panel's heading shows how many are in hand (or not needed) out of the total. It **never blocks and never alerts**.

Process 5 — Details and brochure

1. In the brochure panel fill **Title**, **Photos link** (Google Drive folder), **Description** and **Internal notes**; the code must be code-shaped and free.
2. **Generate** posts to the brochure Google Form ("CODE - Title", folder, details, notes) and logs the request.
3. The Mac scheduler builds the PDF, emails the team and writes it into the inventory; about 30 minutes later the board links the card, fills blanks and **freezes the panel**. "Brochure created" is read from the queue, never ticked by hand.
4. **Redo brochure** unlocks it — in practice, the post-shoot brochure (the owner's "v2"). Versions are not labelled.
5. **Brochure v2 reminder.** When a shoot is completed *after* the brochure was made (for shoots from 13 Oct 2026), the panel shows "Shot on … after the brochure was made — it does not have these photos. Make a new version when the new ones are in." It stays until the brochure is redone or **Not needed** is pressed (`broV2Seen` remembers that for this shoot only).

Process 6 — Shoot

Step	Who	What is captured	Effect
Getting there	Agent	Address, map link, who opens it, occupancy (vacant / owner / tenant / construction), best time, access notes	Shown on the run sheet
Book the shoot	Agent	Date, time (30-minute steps, default 10:00), assigned shooter, site contact	Card moves to Shoot scheduled; clash within ±2 h warns
Owner told	Agent	☐ Owner told we are coming	Required for "ready to send"
Reschedule	Agent	New date, reason	Counts "moved N×", resets Owner told
Cancel	Agent	Reason	Card goes back to Details & docs
Shoot done	Shooter	Media captured ticks, **Drive folder with the raw photos**, notes	Card moves to Shoot done, then automatically to Media ready when nothing is missing
Team calendar	System	A 60-minute event "Shoot — title" at the booked time on the calendar of whoever books it, with the shooter invited; the address, property code, who opens it, what to capture, access notes and a link to the card	Only when **Put booked shoots on the team calendar** is on in Settings (**off by default** — it sends real Google invitations). Moving, re-assigning, cancelling, completing or deleting the shoot moves or removes the event; only the person whose calendar holds it can move it (`shootEventId`, `shootEventBy`)

The **brief** says what to capture: Photos, Video, Drone, Floor plan, Virtual tour (default needed: Photos + Floor plan).

Process 7 — Shoot for, edits per outlet and voice-over

1. **Shoot for**: tick the outlets planned — Insta Story, Insta Reel, FB Reel, YouTube, Collab — and per outlet the voice-over: *With V/O* or *V/O Reqd. separately*.
2. **Plan the edit**: outlets, editor, edit day, deliver-by (must be on or after the edit day). Allowed before the shoot.
3. **Start editing**: only once the shoot is done.
4. **Edit done**: only after starting; ticks that outlet as made. Once done, **+ Final video link** (optional) records where the finished video is (Drive, YouTube…); it can be changed or removed, and a re-edit keeps the old link until a new one is pasted. It is never asked for and never required.
5. **Ask for V/O** appears once the cut is being edited (the rough cut). It lands in the voice-over person's **CRM Daily task**; **V/O made ✓** closes it.
6. **Edit again / Clear** moves the done edit to history (up to 50); a mistaken earlier edit can be removed.

FB Reel reuses the Insta Reel cut unless **FB Reel has its own cut** is ticked.

Process 8 — Automatic move to Media ready

A card moves itself from Shoot done to **Media ready** when *all* of these hold: every needed brief item is done, the photos link is set, every planned outlet is made, every cut is done, every separate voice-over is made, and the brochure is created. Documents, listing type and approvals are **not** checked. The board never moves a card back out of Media ready.

Process 9 — Posting plan, posting and the URL

1. **Paste week plan**: the owner's plan text (e.g. `Monday -S- Lux49 - 4Bhk`) becomes rows: a day, a Story, a YouTube item, or a series row.
2. **Schedule** each channel (Insta Story, Insta Reel, Facebook Reel, YouTube) with a date and time. *Gate:* that outlet's edit must be planned.
3. When the time passes the post shows as **Due** — nothing goes live by the clock alone.
4. **Confirm posted**: "went live at" (not in the future) and the **post URL**. *Gate:* the edit must be done. The card then moves forward to **Live**.
5. A live post without a link stays flagged until the URL is added.
6. **99acres** and **3 PIN website** listings have their own Not posted / Posted / N/A status and URL.
7. Posting the same property again creates a **repost** row with its own schedule and links.

The post URL lives on the Posting row (`postTracker.channels.*.url`) and is shown in Posting, on the Editor card and in the Property dashboard's Shoot & Media section.

Field reference — listing card (`listings/{id}`)

Field	Label / where	Values	Filled at	Why it exists
`stageId`, `stageChangedAt/By`, `reached`	Column, days pill	Stage id, time	Every move	Where the property is and how long it has waited
`leadId`	Open in CRM	Lead id	Creation / Link a lead	Ties the owner's conversation to the property
`title`	Card name ✎	Text	Creation, edits, mapping	The name everyone uses; sent with the brochure
`propertyCode`	Code chip, Property ID	Inventory code	**By hand** at mapping	The single link to inventory, Posting and the lead
`location`, `config`, `askingPrice`	One line	Text	Creation / mapping	Basics for matching and the brochure
`sellerName`, `sellerPhone`, `own`	Summary	Text / flags	From the lead, or typed	Owner contact; a typed edit beats the CRM
`listingType`	Chip	normal / exclusive / collab (or unset)	Onboarding	Mandate type; shown on the property's Shoot & Media block; gates nothing yet
`docs`	Listing & documents	document key → have / applied / missing / na	Details lane	Paper chase status; a checklist only
`address`, `mapLink`, `siteContact`, `occupancy`, `bestTime`, `accessNotes`	Getting there	Text / enum	Before the shoot	So the shooter arrives and gets in
`shootAt`, `shootAssignee(Email)`	Shoot	Time / person	Booking	The booking and who shoots
`shootEventId`, `shootEventBy`	–	Google event id; whose calendar holds it	Booking, with the calendar switch on	Keeps the shoot's calendar entry in step; only its owner may move it
`rescheduled`, `rescheduleReason`, `shootCancelled`, `cancelReason`	Shoot	Count / text	Changes	Spot owners who keep postponing
`ownerInformed`	☐ Owner told	Yes/no	Before the shoot	Avoid wasted trips
`ownerApproved`	☐ Owner approved photos / brochure	Yes/no	After media	The *property owner's* sign-off (not Swami's)
`need`, `media`	The brief	photos, video, drone, floorPlan, tour	Plan / shoot done	What to capture and what was captured
`shootDoneAt`, `shootNotes`	Shot date, notes	Time / text	Shoot done	Unlocks editing; records gaps
`photosLink`	Drive photos	URL	Shoot done / brochure	Raw photos folder; also the brochure's folder
`brochureLink`, `brochure{}`, `description`, `internalNotes`	Brochure panel	URL / state / text	Details lane	Brochure request, lock and source text
`broV2Seen`	Brochure v2 reminder → Not needed	The shoot-done time it was dismissed for	After a post-brochure shoot	Silences the "make a new version" reminder for that shoot only
`fetched`	–	Field → value	Mapping	Exact undo of a mapping

Field	Label / where	Values	Filled at	Why it exists
`forPlan`, `forDone`	Shoot for	Per outlet yes/no	Planning / edit done	Which outlets are planned and made
`forVoice`, `forVo`, `forVoAsk`	Voice over	live / vo; made; ask {at, by, note}	Planning / rough cut	Voice-over per outlet
`cuts`	Edit rows, Editor	planned / active / done + editor, dates, log	Editing	One edit per outlet, with history
`fbOwnCut`	☐ FB Reel own cut	Yes/no	Planning	Whether FB reuses the Reel
`finalLinks`	Final video	cut → { url, at, by }	After Edit done, optional	Where the finished video is (Drive, YouTube…)
`tags`	Tags	Collab, New Dev., Resale, Plot + configured	Any time	Grouping and filters, shown everywhere
`dropReason`, `holdReason`, `holdUntil`	Why?	Enum / date	Drop or hold	Why it stopped, when to revisit
`kind`, `postingId`	Series	series	Posting	A card that groups several properties

Field reference — Posting row (`postTracker/{id}`)

Field	Meaning
`propertyCode`, `propertyId`	Link to inventory and the board card (unique per non-repost row)
`title`, `reference`	Name; the plan's short name ("Lux49 4Bhk"), shared with the card
`plannedDate`, `weekOf`	From the week plan
`channels.igStory / igReel / fbReel / yt`	`{status: yet / scheduled / live / na, at, url, liveAt, day}`
`acres99`, `website`	`{status: pending / posted / na, url, at}`
`repostOf`	A later round of posts for the same property
`kind`, `members`	Series rows (two or more properties)
`tags`, `note`, `details`, `photosLink`	Notes; photos and details live on the card when there is one

Settings → Configurations (`trackConfig`)

Setting	Default	Effect
Editors	none	Listed first when planning an edit; editors open on their own Editor view
Voice-over by (+ email)	none	Owner of V/O requests; marks them "mine" in the CRM Daily task
Deliver by, when not typed	before the post day	Default delivery date for an edit
"Soon" within N days	2	Editor deadline chip
"series" lines become a series	on	Week-plan parsing
New series gets a board card	on	Creates a series card
Put booked shoots on the team calendar (`shootCalendar`)	**off**	A booked shoot goes on the booker's calendar with the shooter invited by Google; moving, re-assigning or cancelling the shoot follows. Sends real invitations — switch on deliberately
Stage targets	2 / 3 / 5 / 2 / 3 / 30 days	Lateness colours and sort
Tags	none	Added to the four fixed tags
Documents checklist (`docItems`)	7 defaults	The documents every card is checked for; add or remove items, **Starting list** restores the defaults; an empty list hides the checklist

3D · Finance — the agency's own books

Console: `3pinfinance.html` (`/3pinfinance`). **Data:** everything under `finance/{tenant}` — settings, `txns` (the ledger), `parties`, `deals`, `invoices`, `bills`, `subscriptions`, `loans`, `assets`, `bankStatements`, `monthEnds`.

Finance keeps 3 PIN Realty's own accrual-basis, double-entry books. Nobody needs to know accounting: the user picks a plain-language action ("Token received", "Invoice the buyer", "Bill received — pay later", "Pay an EMI"), and the app builds a balanced journal against a fixed chart of 63 accounts, plus any bills, invoices, deals, recurring costs, loans or assets it implies — all in one transaction with gapless numbering. Every report is derived live from the ledger.

Screens

Group	Screen	What it is for
Daily	Overview	Month income, costs, profit; **Free to use**; bank, petty, card, loans; who owes whom; GST due; compliance dates; month-end runner
Daily	**Record**	The only screen that writes entries: 7 category tiles (Money in, Money out, Bills, Invoices, Service costs, Deals, Fix something), ~35 actions, live journal preview "Balanced ✓", attach a photo or PDF, Undo within 30 min
Daily	This month	Paid / invoiced / due / late / estimated, "Can I spend?", 3-month outlook
Daily	Transactions	Every entry with filters; drawer with journal, "where this stands" actions and **Reverse**
Daily	Owed	Clients owe you (ageing 0–30 / 31–60 / 61–90 / 90+), you owe vendors, advances, tokens held
Business	Deals & invoices	Deal pipeline per property and side; invoice list with PDF, Share, Upload my own
Business	Recurring	Services, rent, salaries, utilities: expected vs billed, "Record this month"
Business	Budget	Expected vs actual per account, typed overrides
Money	Bank & card statements	Import CSV/XLSX, auto-match, reconcile
Money	Petty cash, Loans, Assets	Box balance; EMI schedules; depreciation
Tax & reports	GST	Output tax, credit, GSTR-3B by head, ITC register, GSTR-1, 180-day rule
Tax & reports	Reports, Analytics, Books	P&L, cash book, cash flow, break-even; trends; trial balance, balance sheet, ledgers, the 15-test books check, exports for the CA (journal, Tally CSV, JSON)
Setup	Profile, Settings, Guide	Company, GSTIN, bank details, digest recipients; rates, prefixes, TDS; an 8-tab in-app guide

A passcode locks every screen except Record (a convenience lock, not security).

Attachments (the photo or PDF on an entry) go to the finance folder in Google Drive, filed by financial year and transaction. The upload route accepts only images and PDFs, and will publish (make link-readable) or delete only a *file* that sits inside the finance folder — the file, its transaction folder, its year folder, then the root: at most three hops. Anything else in Drive cannot be touched through it. The folder is the `driveFolderId` in the finance settings record.

Process 1 — The deal and brokerage cycle

Step	Action	What the books record
Open a deal	Add a deal	Nickname, optional property (code and name copied from the Property dashboard), property state, seller and/or buyer, expected brokerage per side, expected close month. No journal
Token	Token received	Money in, held as a **client advance** (not income)
Registration	Mark registered	Status registered, date, deed number
Invoice a side	Invoice the buyer / seller	Brokerage income + GST (CGST/SGST or IGST by the **property's** state); the token is applied; the rest is receivable. A numbered invoice is always created
Get paid	Client pays	Allocated to invoices oldest first; short-pay as discount or bank charge; extra held as advance
Settle a token	Apply / refund / keep (forfeit) / move	Forfeit becomes income (GST optional) and can cancel the deal
Deal cost	Deal cost	Borne by the company (expense) or recoverable from the client
Problems	Write off, Absorb, Credit note	Bad debt; absorbed cost; numbered credit note reversing income and GST

"Settled" is worked out, not stored: registered, invoiced, nothing owed, nothing held.

Process 2 — Costs, from estimate to settled

1. **Expense** (paid now) or **Bill received — pay later** (creates a bill due in 30 days). "Which month is this cost for?" books it to the right period.
2. **Bill arrived** adds the vendor's bill number to an accrual and releases parked GST credit.
3. **Pay a bill** — oldest first; TDS withheld; a let-off amount reduces the cost.
4. Other actions: petty-cash vouchers, director paid, salary (gross, net, TDS, PF/ESI), buy an asset, sell/scrap an asset, vendor refund.

Process 3 — Recurring costs, loans, month-end

- **Recurring:** set up a service (monthly or paid upfront); each month **Record this month** (paid / not paid yet / skipped), with the variance and its reason; change, pause or cancel (unused prepaid refunded or lost).
- **Loans:** EMI schedule; **Pay an EMI** splits principal, interest, charges, penalty; prepayment rebuilds the schedule; card-to-EMI conversion.
- **Month-end (by hand):** posts prepaid releases and depreciation; asks for a reconciled bank statement first (override recorded).
- **Bank reconciliation:** import a statement → auto-match within ₹1 and ±3 days → suggested entries for unmatched lines (learned rules, ATM = petty top-up, card bill, open bill, EMI, invoice…) → Save reconciliation.

Process 4 — Fixing mistakes

The ledger is **append-only** (enforced by the database rules). A mistake is fixed with **Reverse**: a mirror entry that unwinds payments, voids bills it created and issues a credit note for any invoice it raised. It is refused once an invoice or bill has payments against it.

Key calculations

Figure	Formula
Free to use	Bank + Petty cash − Card balance − Payables − Client advances
GST due	Output GST − Input credit + Reverse charge
GST split	Same state → CGST + SGST (half each); otherwise IGST. Brokerage follows the **property** state; other income the client's state
TDS	Base (before GST) × rate (194H 2 %, 194J 10 %, 194I 10 %, 194C 2 %) — off by default
EMI	P·r(1+r)^n / ((1+r)^n − 1), r = annual rate / 1200
Depreciation	Cost ÷ life (default 36 months), straight line
Profit before tax	Revenue − direct costs + other income − operating costs − depreciation − interest − exceptional items

Field reference — main records

Record	Key fields	Why
Entry (`txns`)	Number, date, action, narration, journal lines (account, debit, credit, party, deal, method), totals, attachments, reversal links	The single source of truth; rules require debits = credits and a date on or after books start
Party	Name, type (client / vendor / employee / lender / director), phone, email, GSTIN, state	Who money moves with; B2B vs B2C; place of supply
Deal	Nickname, property code + name (copied once), property state, seller, buyer, expected amounts, expected month, status (open / registered / cancelled), deed no.	Brokerage per property and side
Invoice	Kind (brokerage / fee / other / forfeit / credit note), number, dates, party, deal, base, GST split, total, paid, status (open / part / paid / void)	What clients owe; the GST invoice PDF
Bill	Vendor, bill number (blank = accrual), dates, period, account, taxable, GST, TDS, total, paid, status	What we owe vendors
Subscription	Kind, vendor, plan, monthly or upfront, history, per-month charges with variance and reason	Recurring commitments
Loan / Asset	Lender, principal, rate, schedule, paid / cost, life, depreciated	Debt and fixed assets
Bank statement	Account, file, mapping, closing balance, rows with match status	Reconciliation evidence

Links to other modules

- **Property dashboard → Finance:** a one-time, read-only copy of property code and name onto a deal.
- **CRM → Finance: none.** A Won lead creates nothing in Finance; deals are entered by hand. This is deliberate for now (TailorTalk's "converted" fires at site visits, so it must never raise an invoice).
- **Daily digest email** at **08:00 IST**: weekly summary on Mondays, renewal alerts 3 days ahead, monthly P&L with the journal CSV on the 2nd.

3E · Platform, data and automations

The platform is five static consoles on one navigation rail, backed by **Firebase** (sign-in, Firestore database, Storage for finance files) and **12 Vercel serverless functions** (the plan's limit, so several jobs share a route). There is one production tenant, `t_3pinrealty`.

Sign-in, tenants and roles

- Everyone signs in with email and password. Each account carries a **tenant** tag; every database read and write is checked against it.
- **There are no enforced roles.** Every team member can do everything in every console, including Finance and Settings. "Editor", "voice-over person" and "team" are settings that route work and set default views, not permissions.
- People are added with `scripts/add-team-member.js`; the team list used for mentions, calendar and site visits is set with `scripts/set-team.mjs`.
- Signing out clears the browser's local copy of the data.

The scheduled day (IST)

Time	Job	What it does
04:30	TailorTalk nightly sync	Pulls every chat active since the last sync, creates / updates leads, runs the AI on changed leads
08:00	Finance digest	Weekly summary (Mondays), renewals in 3 days, monthly P&L (the 2nd); maintains the ledger snapshot
09:00	Follow-up digest	Needs action, overdue, today, tomorrow, day after, @mentions — by email; then rebuilds the lead views
21:30	EOD dashboard summary	Escalations and the day's metrics by email
Every run on the Mac	Brochure delivery + Inventory sync	Delivers finished brochures, then copies the Inventory sheet into the dashboard
Every 5 min (CRM open)	Live refresh	Safety-net TailorTalk pull and due AI reads

Database (Firestore) — what each collection is

Collection	One record is	Written by	Read by
`properties` (+ notes, internalNotes)	A property	Sheet sync, brochure pipeline, dashboard	Every console
`propertyChanges`	One dashboard edit (append-only)	Dashboard	Dashboard, sheet sync (protection)
`leads` (+ notes, history, tailortalk/state, aiRuns)	An enquiry	CRM, TailorTalk sync, AI, board	CRM, dashboard, board, crons
`leadViews`	`{tenant}__meta` (layout, off switch) and `{tenant}__v{n}_{i}` — ~20 leads packed as rows in one record, each stamped `_w` (read-cost saver: the CRM opens with 39 reads instead of 760)	Every lead writer (its row, in the same batch); the 09:00 rebuild; `scripts/lead-views.mjs`	CRM, dashboard, board
`listings` (+ history)	A Shoot & Media card	Board, seller sync	Board, CRM Daily task, dashboard
`postTracker`	A property's posting row	Posting	Board, dashboard
`pipelines/{tenant}`, `trackPipelines/{tenant}`	CRM columns; board columns	CRM; board	Both, crons
`settings/{tenant}`	Team, digests, AI switch, saved views, `trackConfig`	CRM, board, scripts	All
`finance/{tenant}/…`	The books	Finance	Finance, finance cron
`brochureLog`, `brochureDeliveryLog`	Brochure requests; deliveries	API	Create brochure page
`aiQueue`, `aiState`, `ttState`, `ttSignalEvents`, `ttDeadLetters`	AI and TailorTalk bookkeeping	Server only	Server only

Database rules (Firestore ruleset `132b5f73`, deployed 13 Oct 2026)

Every read and write is checked against the caller's tenant. Points worth knowing:

- **Leads:** a team member may `get` or delete a lead that does not exist (so a page can tell "deleted" from "not yours"), but for TailorTalk ids — `tt_<tenant>_<id>`, which can carry a phone number — that answer is given only for the caller's own tenant's ids, so nobody can test whether a number is another tenant's lead. Listing leads needs the tenant field; an update may never move a lead to another tenant.
- **Lead views:** readable by the tenant's team. A page may create or update a view document only with the keys `tenantId` and `rows`, touching **one row per write** — so no page can rewrite or empty a whole document. The meta record and the daily rebuilds are the server's (Admin SDK).
- **Finance ledger** (`txns`): a new entry must balance (debits = credits), carry at least one line and be dated on or after the books' start date; an entry is never deleted — only `reversedBy` and the reconciliation marks may change. The finance settings record (`finance/{tenant}`, including the Drive folder id) is readable and writable by any team member.
- **Property change log** (`propertyChanges`): append-only.

Look and feel (UI v2)

Every console shares `shared-assets/theme.css`: the Inter typeface (served from `shared-assets/fonts/`, OFL licence), neutral greys, one accent (#FF8A00), 11 px minimum text and 24 px minimum tap targets, light and dark. `npm run audit:ui-all` checks every page at desktop and phone width in both themes.

Integrations

Service	Used for	Direction
TailorTalk ("AIRA")	WhatsApp, Instagram and website AI agent; source of most leads	Webhook in; API pull; widget on the home page
Google Sheets	Inventory master (property facts); Queue (brochure intake)	Read by Vercel and the Mac; the Mac writes delivery results back
Google Forms	Brochure intake	Browser → form → Queue sheet
Google Drive	Photo folders, brochure PDFs, finance attachments	Upload via API; PDFs made link-readable
Google Calendar	Site visits, team meetings, shoots (only with the board's Settings switch on; off by default)	Two-way: invitations out, RSVPs back; repeating meetings handled one occurrence at a time
Google Maps	Map, nearby places, travel times	Browser
Gemini 3.5 Flash-Lite	Default lead-automation model	Server → Google
Claude Haiku 4.5	Lead summaries, lost-reason phrasing, alternative lead-automation model	Server → Anthropic
Claude Cowork (on the Mac)	Builds the brochure PDFs	Outside this repo
Gmail (SMTP)	Every outbound email	Server → Gmail
Meta	Optional WhatsApp digest delivery; data-deletion callback	Server → Meta

Release process

The last release was **13 Oct 2026** (`main` at `b48bfaa`, Firestore ruleset `132b5f73`); see *Release status* for what it carried.

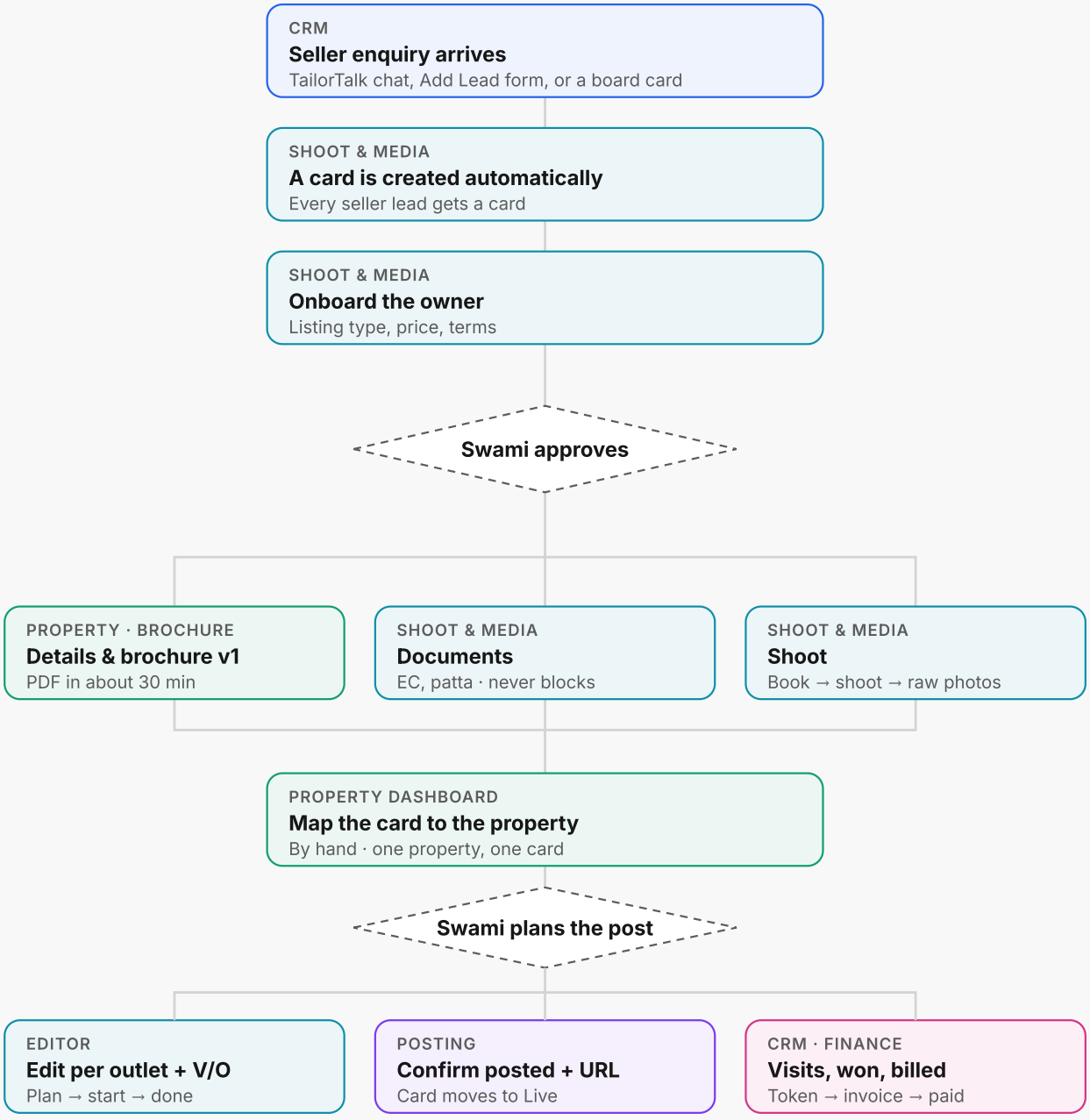
1. Review → fix → re-review until clean; update every caller and test double together.
2. `npm test` (41 suites), the Playwright previews, and `npm run audit:ui-all` at desktop and phone width, light and dark.
3. **Deploy Firestore rules first** (`scripts/rules-diff.mjs`, then `rules-deploy.mjs`, or `deploy-firestore-rules.js`).
4. **Push** `main` — Vercel deploys automatically (no build step).
5. Confirm the live files match the commit, then run the signed-in live smoke test.
6. If the release touched how leads are written: `node scripts/lead-views.mjs verify` must print "IN STEP". `node scripts/lead-views.mjs off` makes every page read the leads directly again, within seconds.

DOCUMENT 4

Architecture & flows

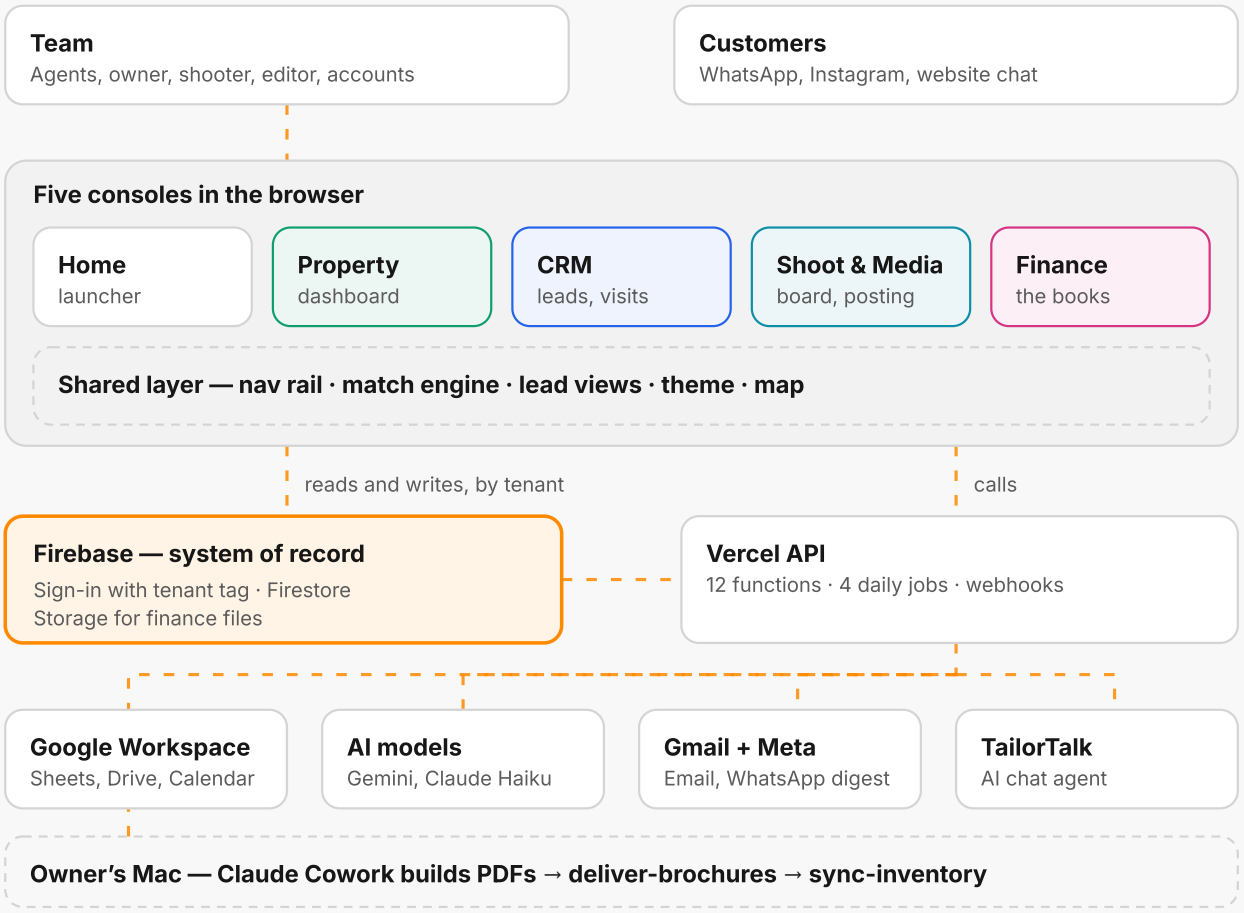
Four animated diagrams of how the platform fits together.

4.1 End-to-end property lifecycle



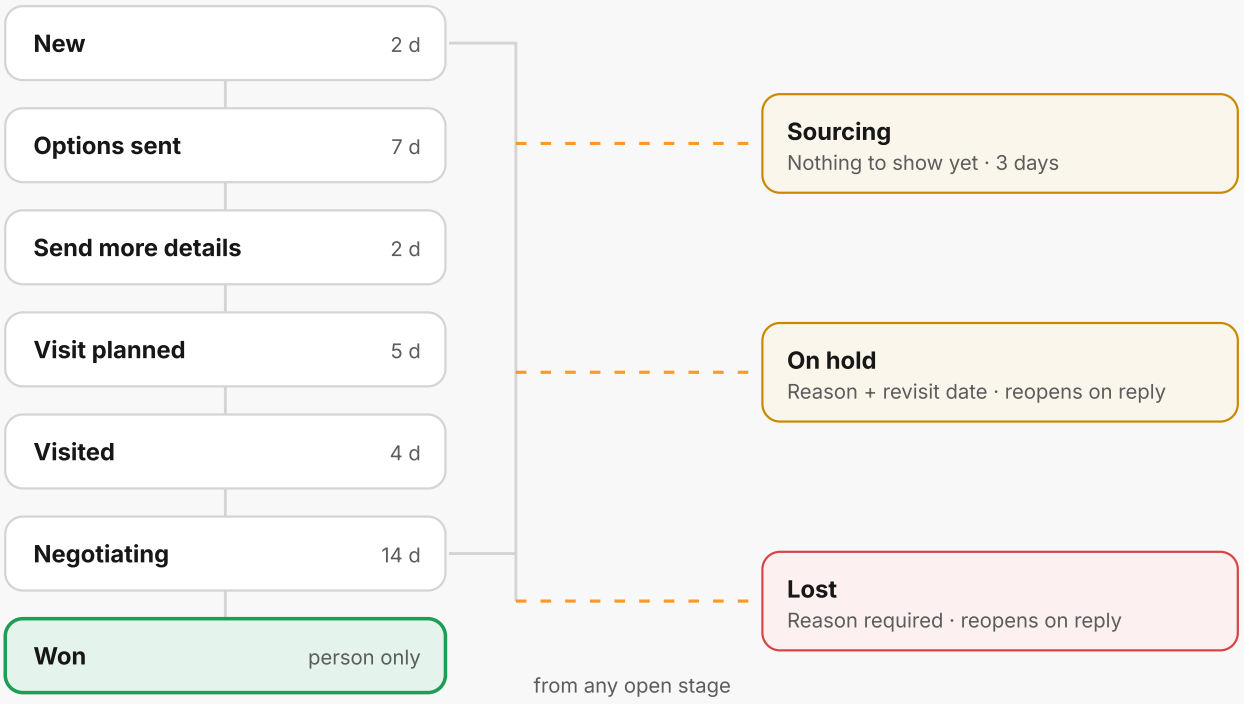
Five consoles, three parallel lanes, two owner gates (dashed) that the app does not enforce yet.

4.2 Interconnection architecture



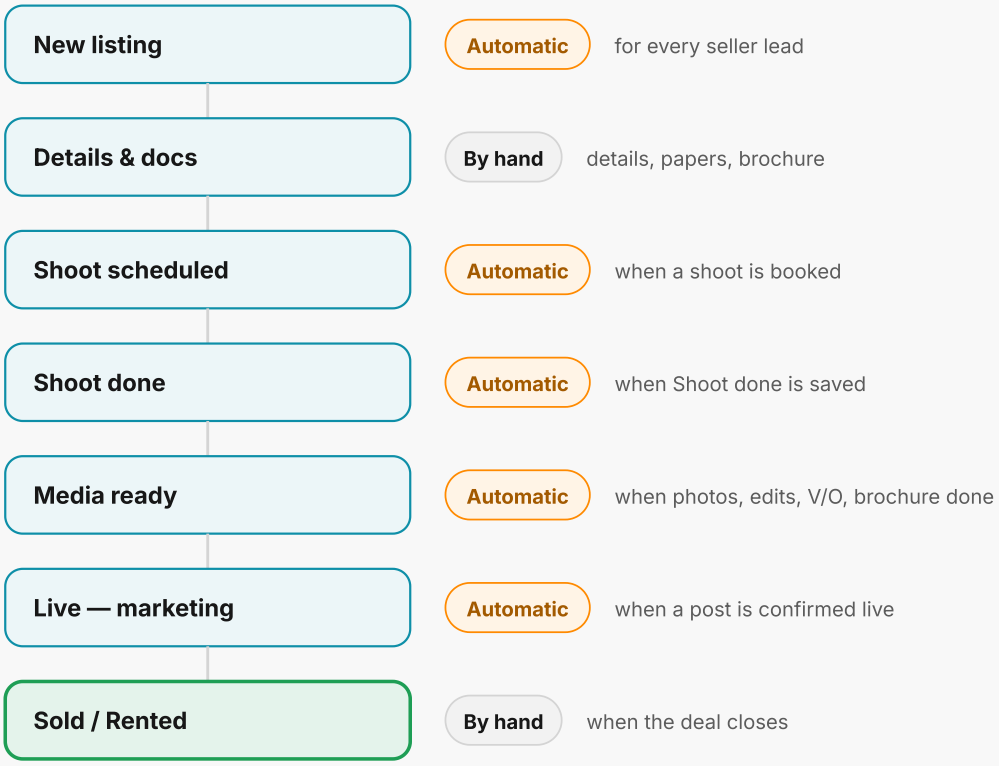
Consoles talk to Firestore directly under tenant rules; anything needing a secret goes through the Vercel API; brochures are built on the owner's Mac.

4.3 CRM lead flow



Seven milestones and three side columns. The AI moves forward only, never to Won, never over a person's choice.

4.4 Shoot & Media card flow



Five of seven moves happen by themselves; On hold and Dropped are always by hand with a reason.

RELEASE

Release status (live 13 Oct 2026), gaps and known issues

What went live, what is still to do by hand, what is in progress, and where the app differs from the owner’s process.

Live is the deployed `main` at commit `b48bfaa` (13 Oct 2026, 22:48 IST) with Firestore ruleset `132b5f73`. Everything the documents above describe is live unless it is marked here as *by hand* or *in progress*. The lead views were built and verified **IN STEP** after the release.

What went live on 13 Oct 2026

Module	Change	Process it affects
Shoot & Media	**Reference on cards** ("Ref: Lux49 4Bhk"), editable on the card	Posting plan
Shoot & Media	**Create a CRM lead from a card** (a duplicate phone offers the existing lead)	Owner onboarding
Shoot & Media	**Move to…** menu on every card; a card can be dropped anywhere on a column	Every stage move
Shoot & Media	**Listing type** Normal / Exclusive / Collab (`listingType`)	Onboarding
Shoot & Media	**Documents checklist** (`docs`; 7 defaults, the list in Settings → Configurations as `docItems`)	Documents lane
Shoot & Media	**Final video link** per cut (`finalLinks`), optional	Final media
Shoot & Media	**Brochure v2 reminder** when the shoot came after the brochure (`broV2Seen` dismisses it)	Brochure
Shoot & Media	**Shoots on the team calendar** — Settings switch `shootCalendar`, **off by default**; sends real Google invitations; `shootEventId` / `shootEventBy`	Shoot scheduling
Shoot & Media	One card per lead; typing-safe live updates; live inventory watch	Mapping, editing
CRM	Keeps scroll; note drafts stay with their lead; follow-ups moved by the AI or a teammate are kept; saves send only changed fields; "⋯" menu on phones	Daily work
CRM	Search and filters hidden on Daily task, Team calendar and Analytics; **Leads for property** by id or code; a seller's **Listing** row opens its card	Navigation
CRM	Team calendar: occurrences of a repeating meeting recognised — the edit form says "Repeats — this change is for this one only"; a single occurrence may be dragged	Meetings
All consoles	**Back** closes the top dialog, then the panel, never the page; a lead or card opened from a link is not reopened once closed	Navigation
All consoles	Saves from a device's old local copy are refused until the server's list has arrived ("Still connecting"): board moves, stage changes and dialogs; dashboard saves except the partial map-pin save	Data safety
Property dashboard	**"Shoot & Media and posts"** block on every property; delete says which cards and leads still point at it; a failed delete is undone and noted in the change log	Property detail
All	**Lead views** (`leadViews`) — the CRM opens with **39 reads instead of 760**; every writer writes the lead's row in the same batch; rebuilt daily at 09:00 after the digest has marked its day as sent	Cost and speed
All	**UI v2**: Inter, neutral greys, #FF8A00 accent, 11 px minimum text, 24 px tap targets (`shared-assets/theme.css`)	Every screen
Finance	**Attachments guard**: images and PDFs only; publish or delete only a file inside the finance Drive folder (at most three hops to the root)	Attachments
Platform	Data-deletion status page escapes everything it prints; TailorTalk safety-net pull resumes where it stopped	Compliance, intake
Database rules	`leads` read split into `get` and `list`, with the "does not exist" answer limited to the caller's own `tt_<tenant>_` ids; new `leadViews` rules (one row per write, keys `tenantId` + `rows` only)	Security

Still to do by hand

Item	Why	How
Lead-views index exemption	Nothing queries inside a view document, and a busy one could otherwise reach Firestore's 40,000 index entries and refuse writes	`node scripts/lead-views.mjs exempt` answered **403**, so set it in the Firebase Console: Firestore → Indexes → Single field → add an exemption for collection `leadViews`, field `rows`, with every index off

In progress — not live

The working copy carries further changes made after the release (CRM, board, dashboard and finance assets, `firestore.rules`, the finance digest, the lead-views client and its test — see `git status`). None of it is deployed; this documentation describes none of it. The same release steps apply when it is ready: review → fix → re-review, `npm test`, previews and `npm run audit:ui-all`, deploy the rules, push `main`, verify the lead views.

Gaps against the owner's lifecycle

Owner's step	Today	Gap
Find on 99acres / WhatsApp / by agent → shortlist	Cards come from CRM seller leads or by hand	No "where found" field or shortlist stage
Gate: Swami approves before any agreement	Not recorded anywhere	No approval step; "Owner approved" means the *property owner* approved the photos
Agreement for Exclusive / Collab before the shoot	Listing type recorded, gates nothing	No agreement fields or shoot gate
Documents: non-blocking, **frequent alerts**	Checklist on every card, non-blocking	No alerts or Daily-task items for missing papers
Brochure v1 and v2	Redo after the shoot; a v2 reminder on the card	Versions not labelled
Gate: Swami plans the post (plan + V/O) before editing	Anyone can plan; editing can start without a post plan	No approver or gate
Final media to Drive	Final link optional	Not a required step
Save the URL against the property	Saved on the Posting row, shown on the dashboard's Shoot & Media block	Not on the lead

Where the app is arguably better than the reference: per-outlet edits and V/O decisions; "posted" confirmed by a person with the URL; mapping by hand with one property per card; a card for every seller automatically.

Known issues in the live code

Area	Issue	Effect	Risk
Shoot & Media	A removed final-video link comes back	The card is saved whole with a merge, so a deleted map key survives; the link reappears on the next refresh	Low
Shoot & Media	Deleting a card leaves its code on the lead	The lead keeps `listingCode`; the seller still looks linked	Low
CRM	AI-written visit dates are not sent to Google Calendar	Invitation keeps the old time until a person re-saves	Medium
CRM	A deleted TailorTalk lead returns on the next sync	Delete is not permanent	Low
CRM	Manage Stages reorder does not persist; a new custom stage jumps to second	Column order	Low
CRM	Matching ignores TailorTalk's requirement text	Weaker buyer matches	Low
CRM	AI Bot editor and WhatsApp digest are largely dead	Use the email digest; replies come from TailorTalk	Low
CRM	Edit dialog → Lost skips the reason	Reason recorded as "other"	Low
Lead views	A lead saved while this browser cannot confirm the view layout is written without its row	The lead is correct; it is missing from lists read through the views until the 09:00 rebuild (or `scripts/lead-views.mjs build`)	Low
Lead views	The rules do not check that a row's `_w` stamp is the server's time	A page could write a wrongly stamped row; the daily rebuild's "keep newer rows" guard then trusts it until the next rebuild from the leads	Low
Property	Toggles re-save the whole property from the page	A stale tab can re-write sheet fields without a log entry (the "Still connecting" guard covers only the first load)	Medium
Property	Deleting a property leaves its notes behind	Orphaned notes	Low
Property	Code patterns differ between board and CRM	Some codes never auto-link to leads	Low
Finance	Reconciled flag usually fails silently; month-end re-run errors	Month-end needs the override	Medium
Finance	Party merge blocked by rules; bank accounts all share id 1000	Duplicate parties stay; one bank ledger	Low
Finance	Opening balances screen probably unreachable	Verify before go-live of a new year	Low
Finance	The finance settings record, including the Drive folder id (`driveFolderId`) the attachments guard trusts, is writable by every team member	A team member could point attachments at another folder	Low
Platform	No enforced roles; Finance passcode is in the page code	Everyone can do everything	Medium
Platform	Team list, default agent, TailorTalk workspace, lead-views switch have no settings screen	Set by script or by hand	Low
Platform	Docs describe removed WhatsApp bot routes; privacy policy is a draft	Documentation and compliance	Low

Fixed in this release and removed from the list: the shoot calendar call now sends the time and a 60-minute length; the data-deletion status page escapes its output; the `leads` rule no longer reveals other tenants' ids.